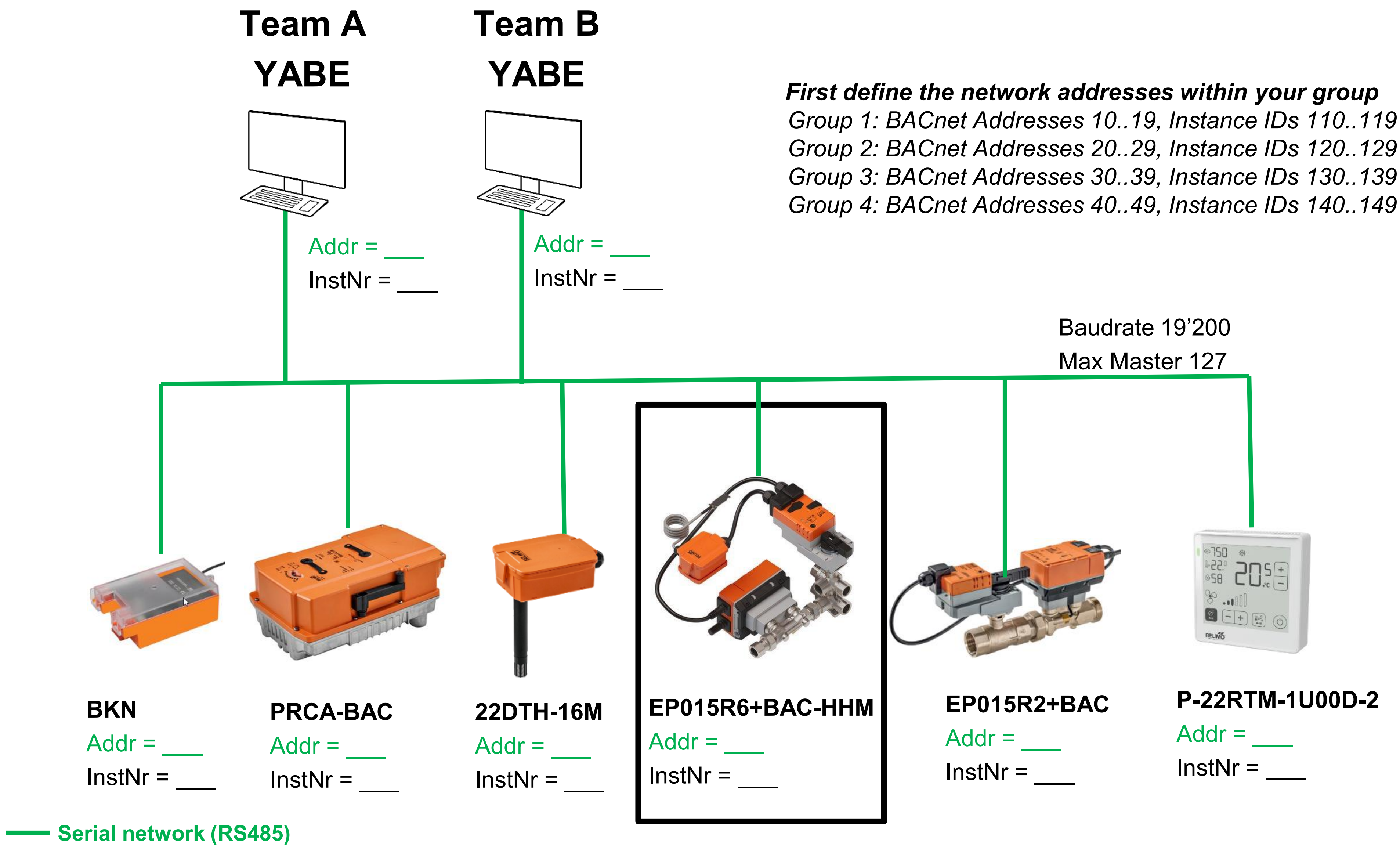
B2 - BACnet / Modbus (Advanced)
BACnet MS/TP - Workshop I

6-way EPIV4

B2 - BACnet - Workshop I

6-way EPIV4

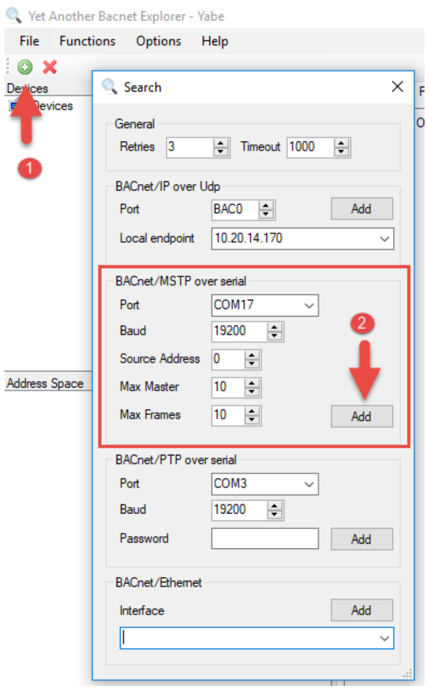


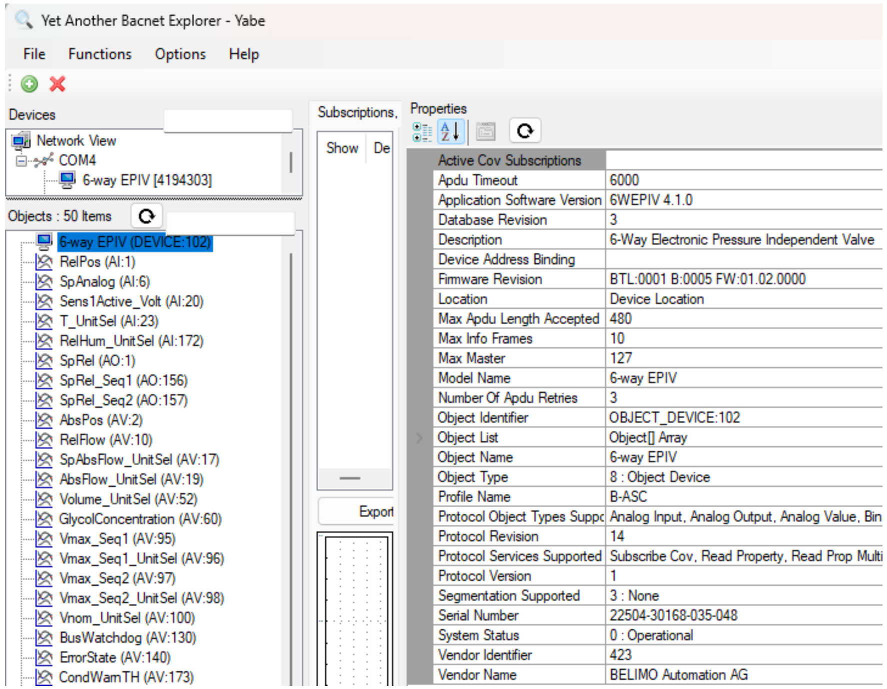
B2 - BACnet - Workshop I

6-way EPIV4

1. **Configure all the devices according the network overview.**
   Find your appropriate configuration tool in the documents (LINK.10, Belimo Assistant 2, etc.)

2. **Start YABE. Click on *Add* device and *Add* yourself with BACnet/MSTP.**
   Check on which COM port your RS485-USB adapter is
   Pay attention to the BACnet settings on your actuator you just made





B2 - BACnet - Workshop I
6-way EPIV4

3. **Select the 6-way EPIV4 under *Devices*.**
4. **Read out all the BACnet objects of the device.**
5. **Click on the BACnet object "Device"**
   1. What is the *Application Software Version*?
   2. With which BACnet object(s) can you write the setpoint(s)?
   3. With which BACnet object can you read out the relative position?

1.

2.

3.

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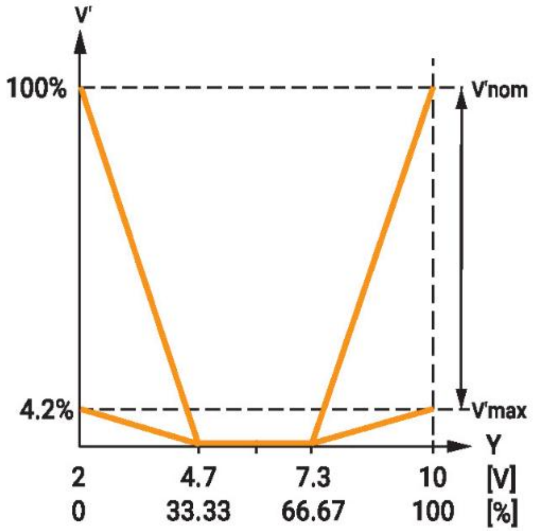
6. **Try as many COV subscriptions as possible. How many COV subscriptions could you make?**

B2 - BACnet - Workshop I

6-way EPIV4

7. Please check out the *mode of operation* in the datasheet.
1. Which setpoint range is within Sequence 1?
2. Which setpoint range is within Sequence 2?



1.

2.

B2 - BACnet - Workshop I
6-way EPIV4

8. **Which *Present Value* of "SpRel" I have to write in order to get Vmax of sequence 1?**

9. **Which *Present Value* of "SpRel" I have to write in order to get Vmax of sequence 2?**

10. **Write the smallest possible "Vmax1" and "Vmax2".**
    1. What is the minimal Vmax?
    2. What happens if I write a smaller value?

1.

2.

11. **Set "Vmax1" and "Vmax2" back to 100%.**

B2 - BACnet - Workshop I
6-way EPIV4

12. **Set Setpoint source to "Bus" and control mode to "Position control".**

Set Setpoint mode BV[155] to "Separate setpoints".

Write 50 % to Setpoint Sequence1 AO[156].

Observe relative position AI[1] on trend using COV subscription.

Actual relative position value:

Write also 50 % to Setpoint Sequence2 AO[157].

Observe relative position AI[1] on trend using COV subscription.

Actual relative position value:

Read error state AV[140]

Value: **Interpretation:**

Set setpoint mode BV[155] back to "Single setpoint".

B2 - BACnet - Workshop I
6-way EPIV4

13. Set a setpoint AO[1] of 100%.

Observe relative position AI[1] change.

Set condensation protection action MV[171] to "Close Valve" .

Read relative humidity AI[172].

Set Condensation alarm threshold AV[174] to a lower value than relative humidity, e.g., *Rel. Humidity: 40% , Alarm Threshold: 35 % .*

Observe relative position AI[1] change.

Read condensation protection status MV[108].

Value: **Interpretation:**

Set Condensation alarm threshold AV[174] to a higher value than relative humidity, e.g., *90%.*

Disconnect the Wago terminal from humidity sensor.

Read condensation protection status MV[108].

Value: **Interpretation:**
